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1.

A pure silicon sample maintained at room temperature has an intrinsic carrier concentration of $1.5 \times 10^{10} \text{ cm}^{-3}$. It is first doped with donors of concentration $2 \times 10^{14} \text{ cm}^{-3}$, followed by acceptors of concentration $4 \times 10^{14} \text{ cm}^{-3}$. Assuming that the carrier mobilities are $\mu_n = 1350 \text{ cm}^2/\text{Vs}$ and $\mu_p = 480 \text{ cm}^2/\text{Vs}$,

- calculate the majority and minority carrier concentrations,
- what is the resistivity of the pure sample, prior to the two types of dopings?
- how will the resistivity change after the dopings?

$$[2 \times 10^{14} \text{ cm}^{-3}, 1.125 \times 10^6 \text{ cm}^{-3}, 2 \times 10^5 \Omega\text{-cm}, 65 \Omega\text{-cm}]$$

$$2. \quad n_i = 1.5 \times 10^{10} \text{ cm}^{-3} \quad N_D = 2 \times 10^{14} \text{ cm}^{-3} \quad N_A = 4 \times 10^{14} \text{ cm}^{-3}$$

$$\mu_n = 1350 \text{ cm}^2/\text{Vs}, \quad \mu_p = 480 \text{ cm}^2/\text{Vs}$$

$$a) \quad N_A - N_D = 2 \times 10^{14} \text{ cm}^{-3} > 0 \Rightarrow p\text{-type}$$

$$\text{major, holes: } p \approx N_A - N_D = 2 \times 10^{14} \text{ cm}^{-3}$$

$$\text{minor, electrons: } n = \frac{n_i^2}{p} \approx 1.125 \times 10^6 \text{ cm}^{-3}$$

$$b) \quad \text{pure sample: } n = p = n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$$

$$\sigma_i = q(n_i \mu_n + n_i \mu_p) \approx 4.392 \times 10^{-6} (\Omega\text{-cm})^{-1}$$

$$\rho_i = \frac{1}{\sigma_i} \approx 2 \times 10^5 \Omega\text{-cm}$$

$$c) \quad p\text{-type: } \sigma_p = q \cdot p \cdot \mu_p \approx 1.536 \times 10^{-2} (\Omega\text{-cm})^{-1}$$

$$\rho_p = \frac{1}{\sigma_p} \approx 65 \Omega\text{-cm}$$

2.

The electron concentration in silicon at $T = 300 \text{ K}$ is given by

$$n(x) = 10^{16} \exp\left(\frac{-x}{18}\right) \text{ cm}^{-3}$$

where x is measured in μm and is limited to $0 \leq x \leq 25 \mu\text{m}$. The electron diffusion coefficient is $D_n = 25 \text{ cm}^2/\text{s}$ and the electron mobility is $\mu_n = 960 \text{ cm}^2/\text{V-s}$. The total electron current density through the semiconductor is constant and equal to $J_n = -40 \text{ A/cm}^2$. The electron current has both diffusion and drift current components. Determine the electric field as a function of x which must exist in the semiconductor.

$$E = 14.5 - (26.0) \exp\left(\frac{+x}{18}\right)$$

$$T = 300 \text{ K}$$

$$n(x) = 10^{16} \cdot \exp\left(-\frac{x}{18}\right) \text{ cm}^{-3}, \quad 0 \leq x \leq 25 \mu\text{m}$$

$$D_n = 25 \text{ cm}^2/\text{s}, \quad \mu_n = 960 \text{ cm}^2/\text{Vs}, \quad J_n = -40 \text{ A/cm}^2$$

$$\Rightarrow \left. \begin{aligned} J_{n, \text{drift}} &= q \cdot \mu_n \cdot n(x) \cdot E(x) \\ J_{n, \text{diff}} &= q \cdot D_n \cdot \frac{d}{dx} n(x) \end{aligned} \right\} J_n = J_{n, \text{diff}} + J_{n, \text{drift}}$$

$$\Rightarrow \frac{d}{dx} n(x) = 10^{16} \cdot \left(-\frac{1}{18}\right) \cdot \exp\left(-\frac{x}{18}\right) = -\frac{1}{18} \cdot 10^{16} \cdot \exp\left(-\frac{x}{18}\right)$$

$$\text{Solve Equation:} \quad \Rightarrow \frac{dn}{dx} = -\frac{1}{18} n$$

$$J_n = q \cdot \mu_n \cdot n(x) \cdot E(x) + q \cdot D_n \frac{dn(x)}{dx}$$

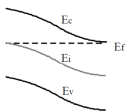
$$J_n + q \cdot D_n \cdot \frac{1}{18} \cdot n(x) = q \cdot \mu_n \cdot n(x) \cdot E(x)$$

$$E(x) = \frac{J_n}{q \cdot \mu_n \cdot n(x)} + \frac{D_n}{18 \cdot \mu_n}$$

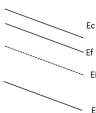
$$\Rightarrow E(x) = \frac{-40}{1.6 \times 10^{-19} \cdot 960 \cdot 10^{16}} \exp\left(\frac{x}{18}\right) + \frac{25}{18 \cdot 960}$$

$$\Rightarrow E(x) = 14.5 - (26.0) \cdot \exp\left(\frac{x}{18}\right) \text{ V/cm}, \quad 0 \leq x \leq 25 \text{ cm}$$

3. For below band diagrams, what are the differences between them? Specify the differences in terms of doping, electrical field, equilibrium etc.



(a)



(b)

① Doping

Both n-type

since diagram (a) is curve $\Rightarrow$ Non-uniform

As for diagram (b) is straight $\Rightarrow$ Uniform

② Electrical field

For (a), exist a built-in non-uniform EF

For (b), exist external uniform EF

③ Equilibrium

For (a), EF is constant $\Rightarrow$ equilibrium

For (b), EF is non-constant $\Rightarrow$ non-equilibrium

4.

4. A p-type silicon sample has an acceptor doping concentration of 10^{16} cm^{-3} . At $t < 0$, it is uniformly irradiated with light of an appropriate wavelength resulting in the generation of electron-hole pairs at a rate of $G_L = 10^{17} \text{ cm}^{-3} \text{ s}^{-1}$. For a temperature of 300 K,

a) assuming a minority carrier lifetime of $\tau_0 = 10 \text{ } \mu\text{s}$, calculate the steady state excess carrier concentrations,

b) determine the quasi-Fermi energy levels and indicate their positions in an energy band diagram.

Take $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$.

$$N_A = 10^{16} \text{ cm}^{-3}$$

$[10^{12} \text{ cm}^{-3}, 0.347 \text{ eV}, 0.109 \text{ eV}]$

$$n_i = 1.5 \times 10^{10} \text{ cm}^{-3}, G_L = 10^{17} \text{ cm}^{-3} \cdot \text{s}^{-1}, \tau_0 = 10 \text{ } \mu\text{s}$$

$$\text{a) } \Delta n = G_L \cdot \tau_0 = 10^{12} \text{ cm}^{-3}$$

$$\text{p-type} \rightarrow \text{major, holes} \Rightarrow \Delta p = \Delta n = \underline{10^{12} \text{ cm}^{-3}}$$

$$\text{b) p-type, } p_0 \approx N_A = 10^{16} \text{ cm}^{-3}$$

$$n_0 = \frac{n_i^2}{p_0} \approx 2.25 \times 10^4 \text{ cm}^{-3}$$

$$\Rightarrow n = n_0 + \Delta n \approx 10^{12} \text{ cm}^{-3}$$

$$p = p_0 + \Delta n \approx 10^{16} \text{ cm}^{-3}$$

$$\Rightarrow E_{Fn} - E_i \approx k \cdot T \cdot \ln\left(\frac{n}{n_i}\right)$$

$$E_i - E_{Fp} \approx k \cdot T \cdot \ln\left(\frac{p}{n_i}\right)$$

$$\Rightarrow E_{Fn} - E_i \approx 0.109 \text{ eV} \quad \text{over } E_i \quad \underline{0.109 \text{ eV}}$$

$$E_i - E_{Fp} \approx 0.347 \text{ eV} \quad \text{below } E_i \quad \underline{0.347 \text{ eV}}$$

5.

5. Consider silicon at $T = 300 \text{ K}$ that is doped with donor impurity atoms to a concentration of $N_d = 5 \times 10^{15} \text{ cm}^{-3}$. The excess carrier lifetime is $2 \times 10^{-7} \text{ s}$. (a) Determine the thermal equilibrium recombination rate of holes. (b) Excess carriers are generated such that $\delta n = \delta p = 10^{14} \text{ cm}^{-3}$. What is the recombination rate of holes for this condition?

[$2.25 \times 10^{11} \text{ cm}^{-3} \text{ s}^{-1}$, $5 \times 10^{20} \text{ cm}^{-3} \text{ s}^{-1}$]

$$T = 300 \text{ K}, \quad N_D = 5 \times 10^{15} \text{ cm}^{-3}, \quad \text{lifetime } 2 \times 10^{-7} \text{ s}$$

$\hookrightarrow \tau_p$

a) Equilibrium $\delta n = \delta p = 0$

$$n_0 \approx N_D = 5 \times 10^{15} \text{ cm}^{-3}$$

$$p_0 = \frac{n_i^2}{n_0} \approx 4.5 \times 10^4 \text{ cm}^{-3}$$

$$R = \frac{p_0}{\tau_p} = \frac{4.5 \times 10^4}{2 \times 10^{-7}} \approx \underline{2.25 \times 10^{11} \text{ cm}^{-3} \text{ s}^{-1}}$$

b) $\delta n = \delta p = 10^{14} \text{ cm}^{-3}$

$$n = n_0 + \delta n \approx 5.1 \times 10^{15} \text{ cm}^{-3}$$

$$p = p_0 + \delta p \approx 10^{14} \text{ cm}^{-3}$$

$$R = \frac{\delta n}{\tau_n} = \frac{\delta p}{\tau_p} \Rightarrow \tau_n \approx \tau_p = 2 \times 10^{-7} \text{ s}$$

$$\Rightarrow R = \underline{5 \times 10^{20} \text{ cm}^{-3} \cdot \text{s}^{-1}}$$

6.

6.

An n-type silicon sample contains a donor concentration of $N_D = 10^{16} \text{ cm}^{-3}$. The minority carrier hole lifetime is found to be $\tau_{p0} = 20 \mu\text{s}$. (a) What is the lifetime of the majority carrier electrons? (b) Determine the thermal-equilibrium generation rate for electrons and holes in this material. (c) Determine the thermal-equilibrium recombination rate for electrons and holes in this material.

[$8.89 \times 10^6 \text{ s}$, $1.125 \times 10^9 \text{ cm}^{-3} \cdot \text{s}^{-1}$]

$$N_D = 10^{16} \text{ cm}^{-3}, \quad \tau_{p0} = 20 \mu\text{s}$$

a) n-type, $n_0 \approx N_D = 10^{16} \text{ cm}^{-3}$

$$p_0 = \frac{n_i^2}{n_0} = 2.25 \times 10^{14} \text{ cm}^{-3}$$

$$\frac{\tau_{n0}}{\tau_{p0}} = \frac{n_0}{p_0} \Rightarrow \tau_{n0} = \frac{n_0}{p_0} \tau_{p0} \approx \underline{8.89 \times 10^6 \text{ s}}$$

b) Equilibrium $G_{th} = R_{th}$

$$G_0 = R_0 = \frac{p_0}{\tau_{p0}} = \frac{n_0}{\tau_{n0}}$$

$$\Rightarrow G_0 = \underline{1.125 \times 10^9 \cdot \text{cm}^{-3} \cdot \text{s}^{-1}}$$

c) $R_0 = G_0 = \underline{1.125 \times 10^9 \text{ cm}^{-3} \cdot \text{s}^{-1}}$

7.

7.

A piece of p-type doped silicon at 300 K has an acceptor concentration of $5 \times 10^{14} \text{ cm}^{-3}$. Assume excess carriers are present and that $E_F - E_{Fp0} = 0.01 k_B T$ and intrinsic concentration of $1.5 \times 10^{10} \text{ cm}^{-3}$.

(a) Does this condition correspond to low injection?

(b) Determine $E_{Fnq} - E_i$.

[Yes, 0.150 eV]

a) $E_F - E_{Fp0} = 0.01 k_B T$

$$N_A = 5 \times 10^{14} \text{ cm}^{-3}, \quad p_0 \approx N_A = 5 \times 10^{14} \text{ cm}^{-3}$$

$$n_0 = \frac{n_i^2}{p_0} \approx 4.5 \times 10^5 \text{ cm}^{-3}$$

$$\Rightarrow \frac{p}{p_0} = e^{\frac{E_F - E_{F0}}{k_B T}} = e^{0.01} \approx 1.01$$

$$\Delta p = p - p_0 \approx 0.01 p_0 \Rightarrow |\Delta p| \ll p_0$$

$$\Rightarrow \text{equilibrium } \Delta n = \Delta p \quad \checkmark \quad p = p_0 + \Delta p$$

$$n = n_0 + \Delta n \approx 5 \times 10^{12} \text{ cm}^{-3}$$

$$\Rightarrow \Delta n \ll p_0$$

Yes, low injection

$$b) T = 300 \text{ K} \quad \checkmark \quad E_{Fp} - E_v \approx kT \cdot \ln\left(\frac{p}{n_i}\right)$$

$$E_{Fn} - E_i = kT \cdot \ln\left(\frac{n}{n_i}\right) \approx \underline{0.150 \text{ eV}}$$

8.

8. A silicon sample at 300 K is n-type with $N_d = 5 \times 10^{16} \text{ cm}^{-3}$ and $N_a = 0$. The sample has a length of 0.1 cm and a cross-sectional area of 10^{-4} cm^2 . A voltage of 5V is applied between the ends of the sample. For $t < 0$, the sample has been illuminated with light, producing an excess-carrier generation rate of $5 \times 10^{21} \text{ cm}^{-3}\text{s}^{-1}$ uniformly throughout the entire silicon. The minority carrier lifetime is $0.3 \mu\text{s}$. At $t = 0$, the light is turned off. Derive the expression for the current in the sample as a function of time $t \geq 0$. Take $\mu_n = 1350 \text{ cm}^2/\text{Vs}$, $\mu_p = 480 \text{ cm}^2/\text{Vs}$ and $n_i = 1.5 \times 10^{10}/\text{cm}^3$

$$T = 300 \text{ K}, \quad N_D = 5 \times 10^{16} \text{ cm}^{-3}, \quad N_A = 0 \quad [54 + 2.2 \exp(-t/\tau_{p0}) \text{ mA}]$$

$$L = 0.1 \text{ cm}, \quad A = 10^{-2} \text{ cm}^2, \quad V = 5 \text{ V}$$

$$t < 0 \quad \text{uniform} \quad G_0 = 5 \times 10^{21} \text{ cm}^{-3} \cdot \text{s}^{-1}$$

$$\text{holes} \quad \tau_{p0} = 0.3 \mu\text{s}$$

$t = 0$ shut down

$$\mu_n = 1350 \text{ cm}^2/\text{Vs}, \quad \mu_p = 480 \text{ cm}^2/\text{Vs}, \quad n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$$

$$\Delta p_0 = G_0 \cdot \tau_{p0} = 1.5 \times 10^{15} \text{ cm}^{-3} \quad \checkmark \quad \Delta n_0 = \Delta p_0$$

$$\Delta p(t) = \Delta p_0 \cdot e^{-\frac{t}{\tau_{p0}}} = 1.5 \times 10^{15} \cdot e^{-\frac{t}{0.3 \mu s}} \Rightarrow \Delta n(t) = \Delta p(t)$$

$$\Rightarrow \sigma(t) = q \cdot [\mu_n(n_0 + \Delta n(t)) + \mu_p(p_0 + \Delta p(t))]$$

$$n_0 = N_D = 5 \times 10^{16} \text{ cm}^{-3}$$

$$p_0 = \frac{n_i^2}{n_0} = 4.5 \times 10^3 \text{ cm}^{-3} \ll \Delta p(t)$$

$$\Rightarrow \sigma(t) \approx q \cdot \mu_n n_0 + q(\mu_n + \mu_p) \Delta p(t) = \sigma_0 + \Delta \sigma(t)$$

$$\hookrightarrow \sigma_0 = q \cdot \mu_n \cdot n_0$$

$$\hookrightarrow \Delta \sigma(t) = q \cdot (\mu_n + \mu_p) \cdot \Delta p(t)$$

$$\Rightarrow \sigma_0 = 10.8 \text{ S/cm}, \quad \Delta \sigma(0) = 0.439$$

$$\Rightarrow E = \frac{V}{L} = 50 \text{ V/cm}, \quad J(t) = \sigma(t) \cdot E$$

$$I(t) = J(t) \cdot A = [\sigma_0 + \Delta \sigma(t)] \cdot E \cdot A$$

$$\Rightarrow I(t) = [10.8 + 0.439 e^{-\frac{t}{\tau_{p0}}}] \cdot 50 \times 10^{-2} \text{ mA}$$

$$\Rightarrow \underline{I(t) = 54 + 2.2 e^{-\frac{t}{\tau_{p0}}} \text{ mA}}$$

9.

Consider a semi-infinite p-type silicon bar doped homogeneously to a value of $1.39 \times 10^{16} \text{ cm}^{-3}$ at 300 K. The intrinsic carrier concentration is $1.5 \times 10^{10} \text{ cm}^{-3}$. The applied electric field is zero. A minority carrier concentration is electrically injected at one end of the sample ($x=0$) such that the excess minority carrier concentration at $x=0$ is $2.49 \times 10^{13} \text{ cm}^{-3}$. The electron mobility and the electron lifetime are $1450 \text{ cm}^2/\text{V}\cdot\text{s}$ and 10^{-6} s respectively.

i) Calculate the steady state excess electron concentration as function of x

ii) Calculate the electron diffusion current density at $x=0$ and $x=10 \mu\text{m}$

$$[2.49 \times 10^{13} \exp(-x/61.4 \times 10^{-4}) \text{ cm}^{-3}, 24.5 \text{ mA/cm}^2, 20.8 \text{ mA/cm}^2]$$

$$N_A = 1.39 \times 10^{16} \text{ cm}^{-3}, \quad T = 300 \text{ K}, \quad n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$$

$$\Delta n(0) = 2.49 \times 10^{13} \text{ cm}^{-3}, \quad \mu_n = 1450 \text{ cm}^2/\text{V}\cdot\text{s}, \quad \tau_n = 1 \mu\text{s}$$

$$a) \quad D_n = \mu_n \cdot \frac{kT}{q} = 37.5 \text{ cm}^2/\text{s}$$

$$L_n = \sqrt{D_n \cdot \tau_n} \approx 61.4 \times 10^{-4} \text{ cm}$$

$$\Delta n(x) = \Delta n(0) \cdot e^{-\frac{x}{L_n}} = 2.49 \times 10^{13} \times \exp\left(-\frac{x}{61.4 \times 10^{-4}}\right) \cdot \text{cm}^{-3}$$

$$b) J_n(x) = q \cdot D_n \cdot \frac{d\Delta n}{dx} = q \cdot D_n \left(-\frac{\Delta n(0)}{L_n}\right) \cdot e^{-\frac{x}{L_n}}$$

$$\Rightarrow J_n(x) = -24.5 \cdot e^{-\frac{x}{L_n}} \text{ mA/cm}^2$$

$$|J_n(0)| = 24.5 \text{ mA/cm}^2$$

$$|J_n(L_n)| = 20.8 \text{ mA/cm}^2$$

10.

10.

Consider a semi-infinite n-type silicon bar doped homogeneously to a value of $2.16 \times 10^{16}/\text{cm}^3$ at 300 K. The intrinsic carrier concentration is $1.5 \times 10^{10}/\text{cm}^3$. The applied electric field is zero. A minority carrier concentration is electrically injected at one end of the sample ($x = 0$) such that the excess minority carrier concentration at $x = 0$ is $1.6 \times 10^{13}/\text{cm}^3$. The hole mobility and the hole lifetime are $450 \text{ cm}^2/\text{V}\cdot\text{s}$ and 10^{-8} s respectively.

- Calculate the steady state excess hole concentration as function of x
- Calculate the hole diffusion current density at $x = 0$.

n-type

$$[1.6 \times 10^{13} \exp(-x/3.42 \times 10^{-4})/\text{cm}^3, 87.6 \text{ mA/cm}^2]$$

$$N_D = 2.16 \times 10^{16} \text{ cm}^{-3}, T = 300 \text{ K}, n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$$

$$\Delta p(0) = 1.6 \times 10^{13} \text{ cm}^{-3}, \mu_p = 450 \text{ cm}^2/\text{Vs}, \tau_p = 10^{-8} \text{ s}$$

$$a) D_p = \mu_p \cdot \frac{kT}{q} = 11.66 \text{ cm}^2/\text{s}$$

$$L_p = \sqrt{D_p \cdot \tau_p} = 3.42 \times 10^{-4} \text{ cm}$$

$$\Rightarrow \Delta p(x) = 1.6 \times 10^{13} \cdot e^{-\frac{x}{3.42 \times 10^{-4}}} \text{ cm}^{-3}$$

$$b) J_p(x) = -q \cdot D_p \cdot \frac{d\Delta p}{dx} = q \cdot D_p \cdot \frac{\Delta p(0)}{L_p} \cdot e^{-\frac{x}{L_p}}$$

$$\Rightarrow J_p(0) = \frac{q \cdot D_p \cdot \Delta p(0)}{L_p} = 87.6 \text{ mA/cm}^2$$